
Aethl
Bio

All Wounds Need Oxygen to Heal. Chronic Wounds are Starved of it.

6.5M
Americans living with chronic wounds

1.5M
Diabetic foot ulcers in the US annually

$50B+
Annual US wound care costs

Wound tissue sits at 0-10 mmHg of oxygen. Healing requires more than 4 times that.

Exposing a wound to ambient air does not close that gap.

All Wounds Need Oxygen to Heal.
Chronic Wounds are Starved of it.

CURRENT DFU TREATMENT

$15k - $40k

HBOT, NPWT, advanced dressings

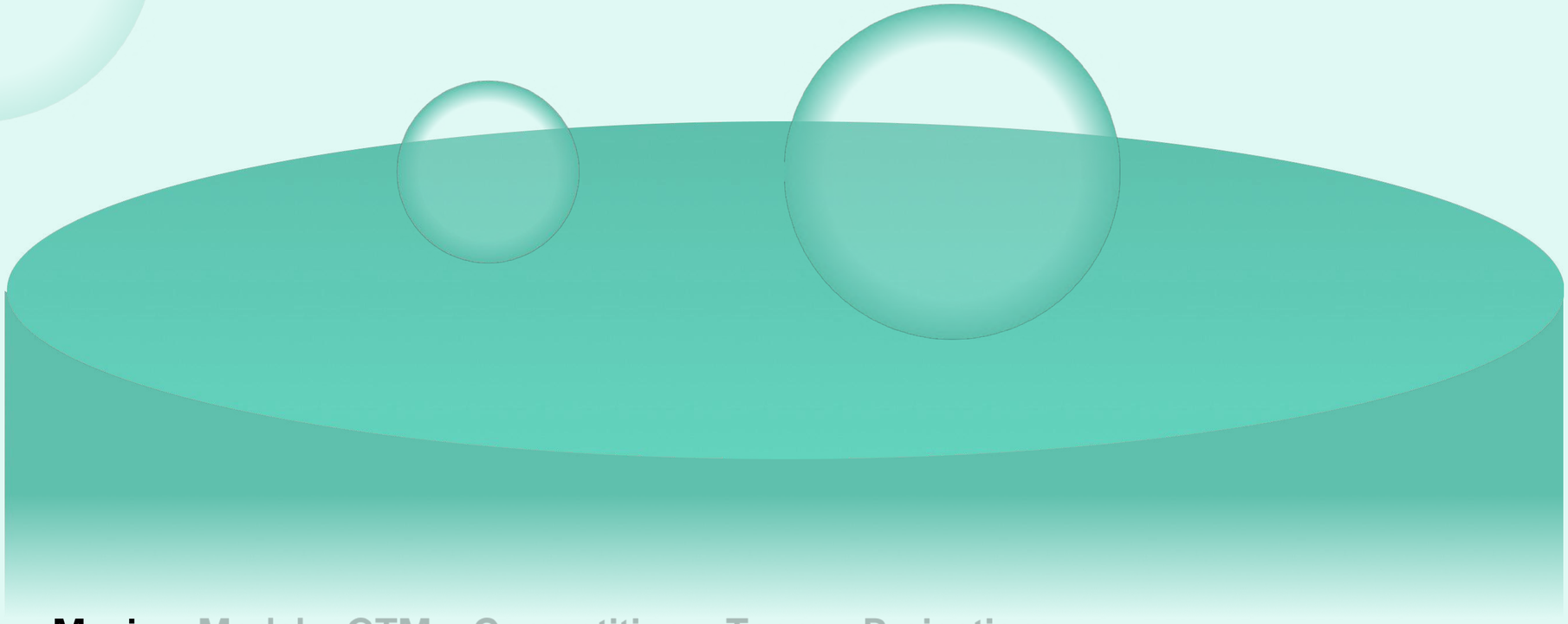
AETHL EXPECTED TREATMENT COST

$0.1k - $5k

Up to 10× lower cost of care*

No chamber, no hospital visit.
Shelf-stable topical oxygenation therapy.
A fraction of the cost.

Aethl will revolutionize healing with a new and economical alternative to oxygen therapy: **suspend it in water**



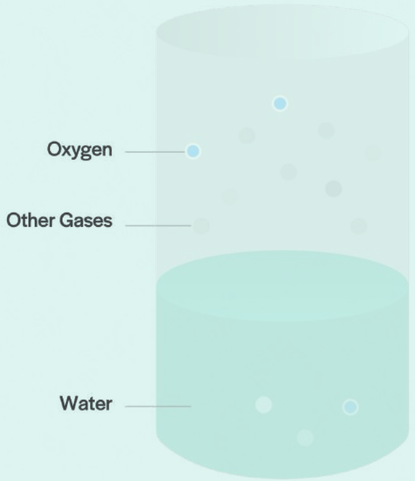
Our Hydro-Activated Supersaturated Solution.

The Platform Revolutionizing Healing.

Dissolved oxygen held in proprietary hydrogel solution at 2,000 mmHg, over 13x ambient conditions air and 3x pure oxygenated water. Aethl's hydrogel solution can carry other compressed gases and small molecules.

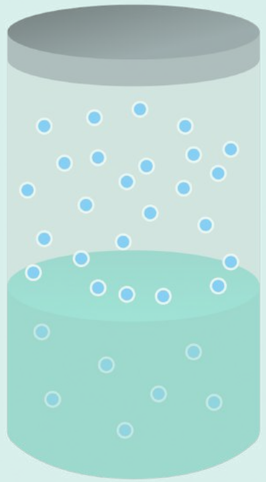
- De-risked from day one
- No license fee or legal barriers
- Our process, products, and clinical use are protected by patents
- Capital efficient investments
- Regulatory and scale moat

157 mmHg



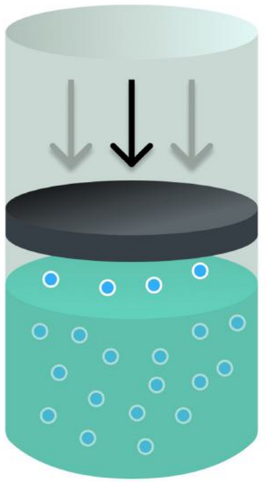
Ambient Air

760 mmHg



Pure O_2, oxygenated water

2000 mmHg



Hyper- oxygenated H_2O

INTRODUCING,

Nuvoxen.



- **Patents pending**
  Process, product, & clinical use
- **Indefinite Storage**
  Shelf-stable pressurized format
- **Scalable**
  Proprietary manufacturing
- **Reimbursable**
  Coded for immediate use (A6248)

A wound that wouldn't close.

Until Aethl. *(Case Study)*

Single-patient compassionate use.

Male, mid-to-late 80s. Stage III plantar pressure ulcer, right foot, first MTP joint. Subject self-applied the prototype gel as a cosmetic. Aethl conducted no recruitment, provided no oversight, and made no clinical claims.

18
months
refractory to treatment before first Gel application

20 - 30
millimeters (mm)
open wound diameter through epidermis into dermis at baseline

- This case was severely recalcitrant
- Standard-of-care closure timelines span weeks-to-months under optimal offloading
- This subject has access to the best options in wound care and lifestyle was severely altered due to this wound

Prototype Gel, 2 - 3× daily, ambient air, no occlusive dressing. **Wound closure at Day 10** ***(not pictured).***



PRESENTATION

Baseline. Open ulcer; exposed granulation; hyperkeratotic periwound. Ten months refractory from last treatment.



PREPARATION

Post-debridement, Gel applied. Aqueous film with visible dissolved-O_2 bubbling at the wound bed.



CLOSING

Wound area reduced ~60 - 75%. Active epithelial migration; healthy granulation; no infection.

No claims, or control: Aethl Bio, Inc. did not recruit, enroll, or direct the subject. The subject applied the gel voluntarily as a commercially available cosmetic product.

Closed; not healed: the subject re-injured the site during ambulation, a predictable consequence of inability to offload a load-bearing surface.

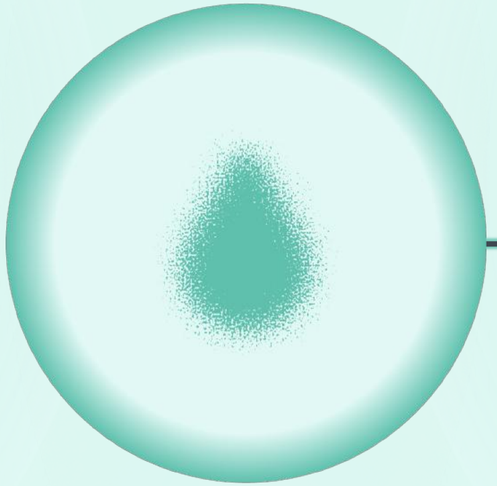
Aethl is biomaterials Platform.

We launch with Oxygen.

We have scalable manufacturing capability that is proven on oxygen. The same IP, the same supply chain, and the same line will produce a pipeline of compressed therapeutic gases and other small molecule hydrogel applications.

FDA agreed our product, Nuvoxen, is a 510(k) product and not a drug.

- FDA confirmed Aethl's overall regulatory strategy is appropriate for clearance.
- Our predicates and biocompatibility testing plan agreed upon.
- No animal data required for submission.



HAS Platform

Oxygen is application one.
It is not the company.

LAUNCH

O_2

Oxygen

Wound healing • OTC wellness

PIPELINE

NO

Nitric Oxide

Anti-inflammatory • analgesia

PIPELINE

CO_2

Carbon dioxide

Vasodilation • microcirculation

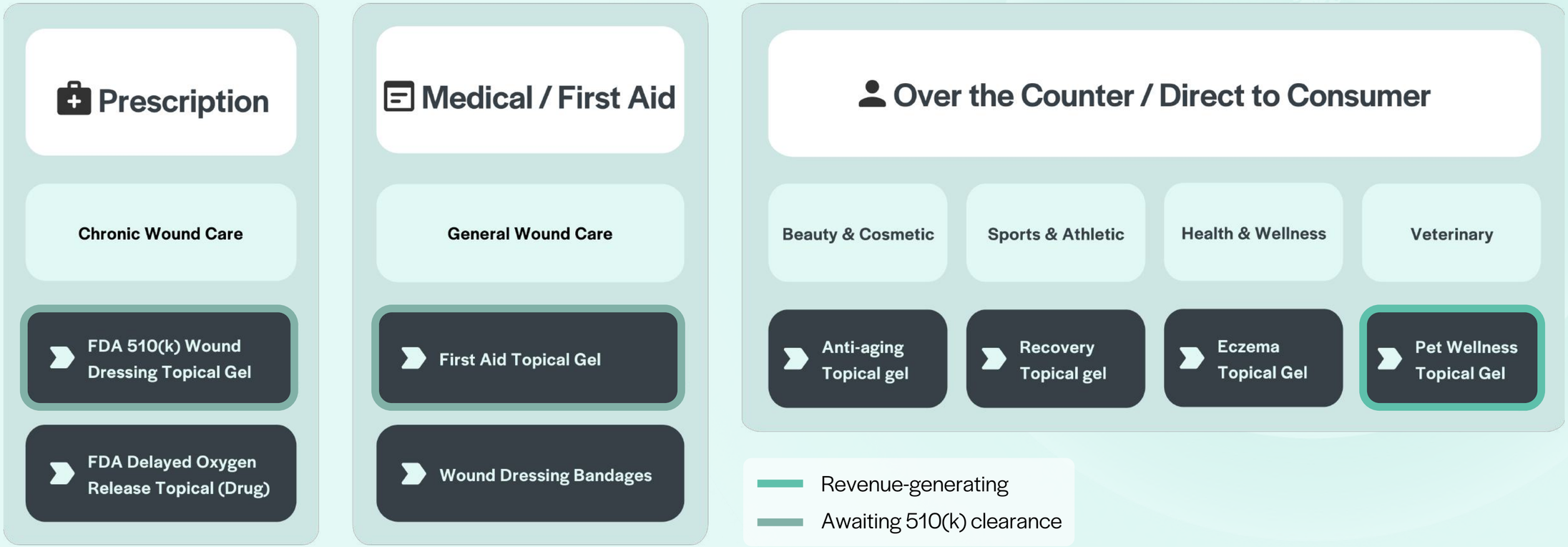
PIPELINE

H_2S

Hydrogen sulfide

Cytoprotection • tissue repair

One Platform.
Three Revenue Engines.



Revenue Now.
Reimbursement Next.

A strategy in parallel derisking our proposition.
D2C cashflow that continue to fund our therapeutic platform.

Initial Customer Acquisition

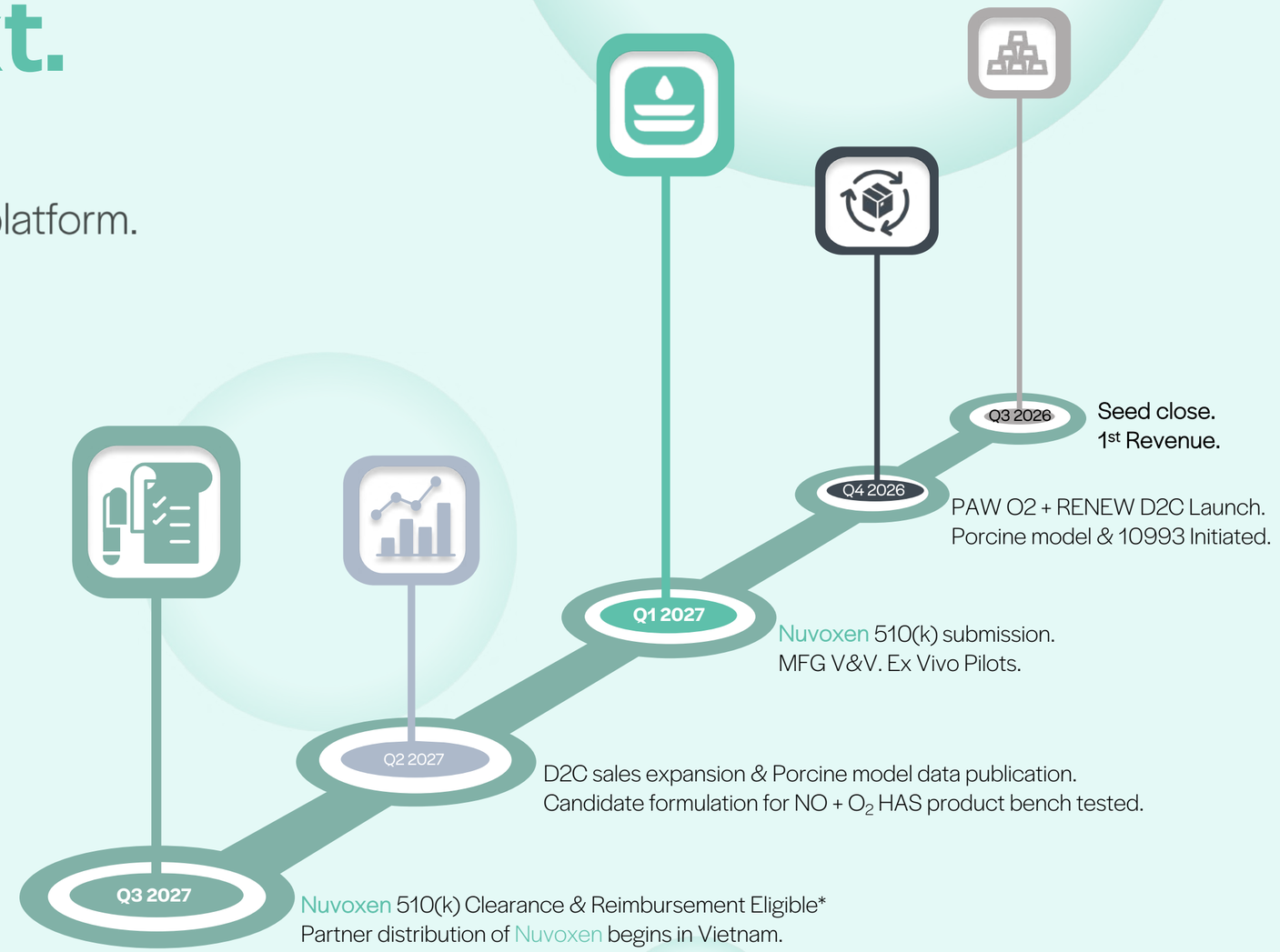
DIRECT TO CONSUMER

- Amazon
- Shopify
- Veterinary
- Social Media

PRESCRIPTION USE

- Wound Care Centers
- Podiatry Networks
- Trade Conferences
- Post-Market Studies

Nuvoxen's reimbursement code is **A6248** (plain hydrogel covering), with pays roughly $23 to $28 per fluid ounce. Our launch SKU of 50ml SKU will gross about $39 to $47 per unit, and our 100ml extension SKU will gross near $69 to $83. Each payment counts for a single wound per month at the 3.4 fluid ounce per wound per 30-days, established by CMS. The A6248 code is just the floor for our product, as we plan to gather data during post market activity to build toward a dedicated code given our formulation advantages.



Signed Purchase Order in Hand.
First Revenue Booked.

$100K

PAW O_2 purchase order

PO# AG-GVH-2026-001 · June 3, 2026
Animal General of Cranberry Twp. / Gardens Veterinary Hospital

✓ No 510(k) required.

The veterinary path needs no FDA clearance . We're generating revenue today while Nuvoxen advances on its own track.

✓ Buyer is a multi-site veterinary network.

A multisite veterinary network (Animal General) in Pennsylvania is purchasing 4000 units for pilot use.

✓ Built to scale.

Pre-authorization pilot and Value Analysis Committee passed. Lays groundwork for repeat sale.



Nuvoxen is a Low-Cost Chronic Wound Dressing Solution for Southeast Asia

The diabetic-foot pathway remains the deadliest, costliest non-healing wound burden in Southeast Asia. In July 2026 we executed an LOI with an established Vietnamese distribution partner for exclusive, three-year distribution of Nuvoxen in Vietnam giving us a path to market-entry, revenue, and profitability. Our executed LOI given our partner's own MOQ implies $4-14M annually at our partner's stated distribution intent. **The agreement gives us validated demand and a playbook for expansion.**

EXECUTED LOI

JULY 2026

Exclusive distribution relationship for Vietnam with binding confidentiality and non-binding terms. Key terms include MOQ, pricing, and order frequency.

NETWORK

≈11,000

Our partner's distribution footprint in Vietnam is country-wide, which includes 11,000+ clinics and 11 hospitals in the network.

SCOPE

3 YEARS

Exclusivity spans Nuvoxen, future medical products, agricultural, and veterinary lines. Aethl retains sole ownership of all data, IP generated.

PATH TO PROFITABILITY

$4M PER YEAR

Modeled revenue at the LOI's minimum order cadence over the three-year term, MOQ volumes carry Aethl to operating profitability in Y1 of the agreement.

PILOT MARKET

VIETNAM

≈ 5.8M
adults with diabetes, of which about half are undiagnosed

≈ 365K
with an active diabetic foot, venous or pressure ulcers at any given time

≈ 40%
health spend out-of-pocket, despite approximately 90% of patients are insured

Q4 2026 : Clinical Feasibility

30 - 60 day program using 100 units assessing wound-closure trajectory, closure rates, tissue-oxygenation, outcomes data, user feedback, and health-economic data.

Agreement finalized by Jan 1, 2027

Leveraging the LOI as a foundation for negotiations, the parties shall execute agreement following the feasibility assessment. Partner purchase decision and first commercial orders follow at MOQ quantities.

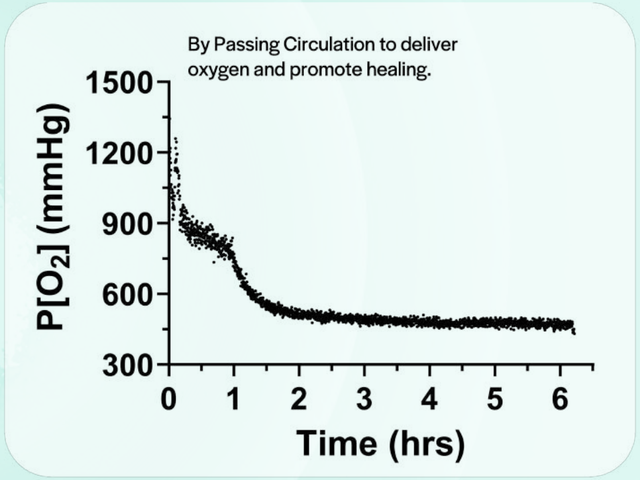
H2 2027: Product in-country

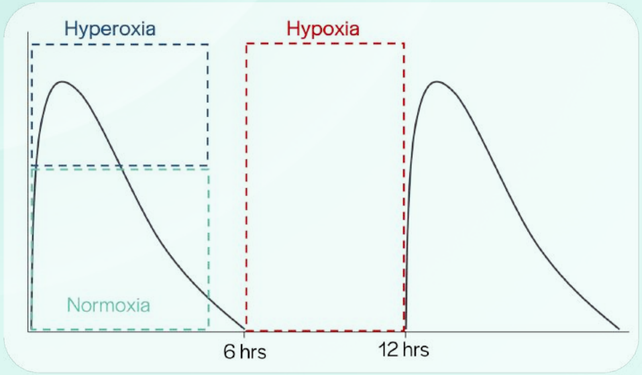
Initial order of units delivered across Vietnamese hospitals and clinics based on the LOI's feasibility conditions and US FDA 510(k) clearance. The order represents an initial six-figure-unit order to be distributed with repeat orders to follow.

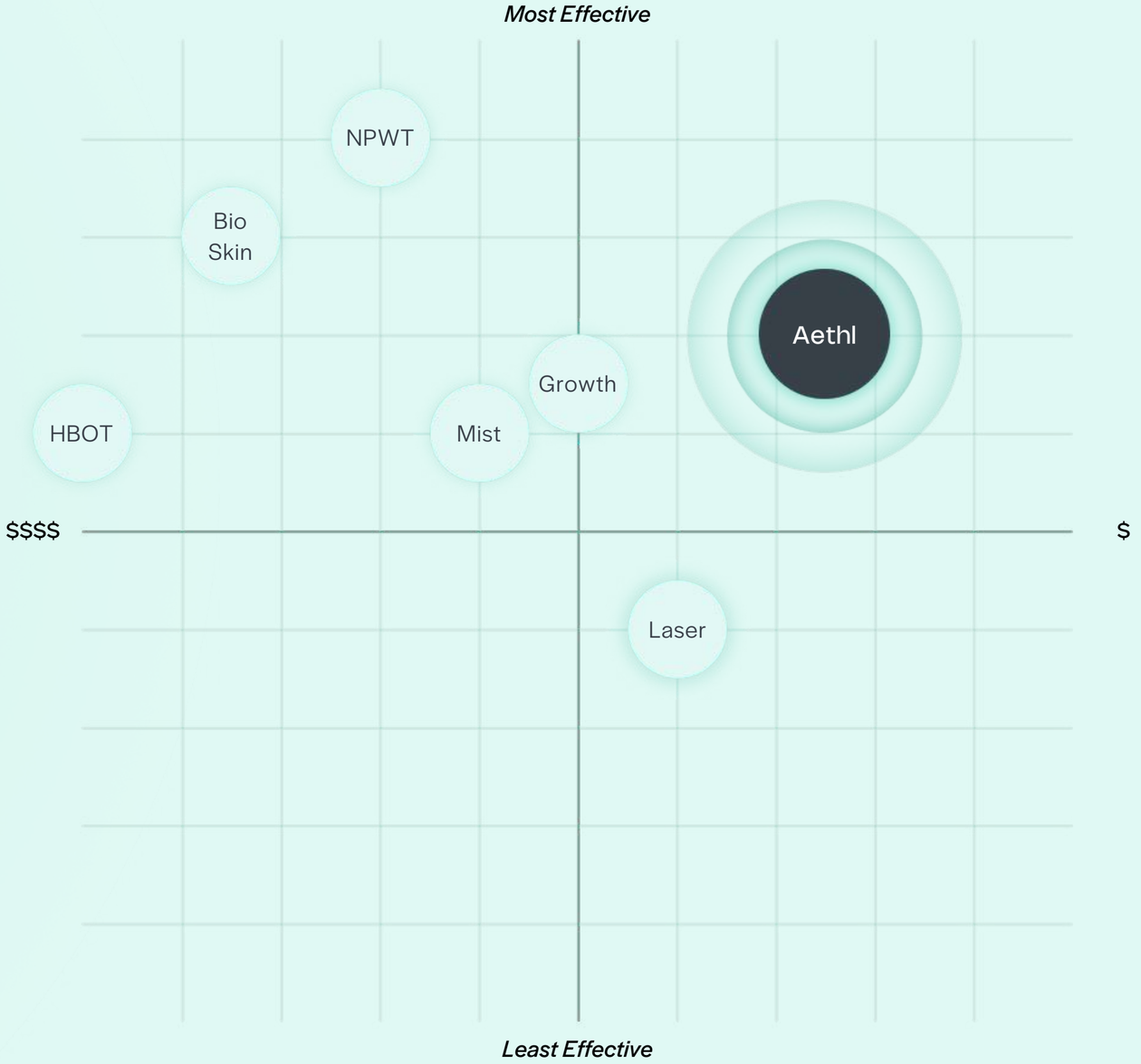
More effective.
Dramatically cheaper.

Aethl's Nuvoxen wound dressing provides hyperbaric-levels of oxygen for 2 hours, which will result in several hours of tissue oxygenation (*right*). With application every 12 hours, our tissue oxygen concentration can be illustrated as the graph above (*bottom right*). Our dressing provides best in market oxygenation dosing, at a fraction of the cost.

This dosing allows for periods of: (1) hyperoxia that engage keratinocytes, epithelial cells, and fibroblasts; (2) normoxia to stabilize the wound environment, and (3) hypoxia to stimulate fibroblast activation and release of angiogenic factors. Maintaining stable oxygen content in the wound site is counterproductive to optimal repair timing.







Aethl Bio

Repeat Founders.
Industry Veterans.



Carmelo R. Montalvo

Chief Executive Officer

An audacious mission-driven, strategy-minded leader and multiple-time founder including a medtech exit, Carmelo has over a decade of experience building, scaling, and optimizing organizations at the intersection of tech, operations, and innovation. Recognized KOL in organizational leadership and known for executing operational transformation and driving excellence in high-growth environments, Carmelo is adept at translating complex challenges into scalable solutions that deliver measurable business value, including a proven track record of successfully navigating the regulatory landscape to achieve multiple FDA clearances.



Dan Willis

Chief Technology Officer

An accomplished engineer and 2x founder, Dan possesses over a decade of experience in electrical and software engineering. He specializes in diagnosing complex field issues and implementing impactful manufacturing process changes for products like gas detection sensors, showcasing a proven ability to deliver tangible results from a deep technical foundation. Dan has a patented track record in translating complex concepts into viable, impactful solutions, demonstrating a firm grasp of the full innovation management lifecycle from initial idea generation through rigorous development in regulated environments and market-ready iteration.



Delvina Morrow

Chief Operating Officer

A tactical operations executive and organizational builder, Delvina brings nearly two decades of leadership spanning life sciences, technology, professional sports, and the social sector. She specializes in constructing the teams, systems, and partnerships that high-growth, mission-driven organizations need to scale, from fundraising infrastructure and stakeholder strategy to brand development and cross-functional operations. Known for translating complex organizational challenges into clear operational architecture, Delvina has a proven track record of designing multimillion-dollar strategies, building from the ground up, and shaping nationally visible engagement platforms.



Dr. Melissa Thompson, PhD

Chief Compliance Officer

A seasoned regulatory and quality leader with over 30 years of experience, Melissa excels at turning life-science concepts into commercialized therapies and medical technologies. Known for a practical, collaborative approach during pivotal moments, she possesses a proven track record navigating complex FDA/EMA landscapes for drugs, biologics, and devices. Adept at resolving regulatory setbacks and building robust quality systems, Melissa thrives on partnering with founders to chart achievable paths to market, combining deep technical expertise with a genuine passion for advancing patient-centered innovation.

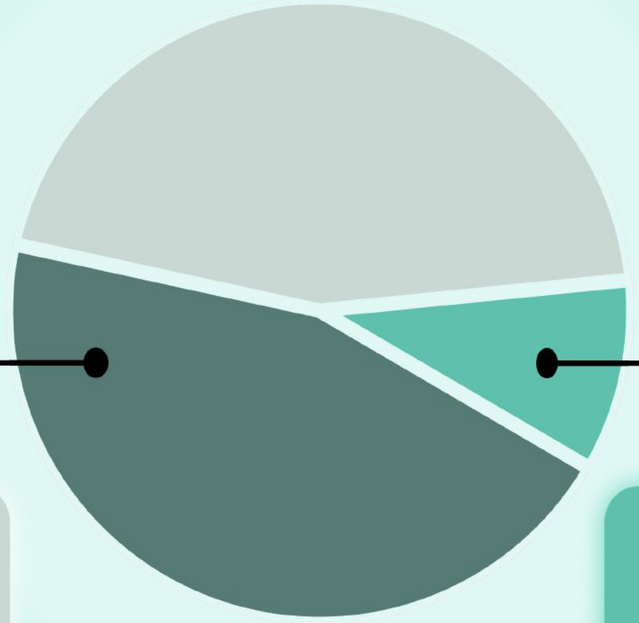
Massive Market.
Better Outcomes.

Diabetic foot ulcers affect about 1.5 million people in the US each year, and current treatment costs $15k to $40k per case. Aethl treats the same condition for $5k to $10k with comparable efficacy, which is roughly 10x less expensive than standard care and four times less expensive than hyperbaric oxygen.

Total Addressable Market

>$50B

1.5M diabetic foot ulcers (DFU) occur annually in the US



Serviceable Addressable Market

$22.5B - $60B

Current DFU treatment cost is $15k-40k

Serviceable Obtainable Market

$.1B - $5B

Aethl DFU treatment cost is $5k-10k

Assuming 1.3% to 33% market penetration

10x

Lower cost of Care vs standard DFU

4x

Cheaper than HBOT with comparable efficacy



Carmelo R. Montalvo
carmelo@aethl.com
412.212.6563
www.aethl.com

Headquarters
100 S Jackson Ave
Ste 05-526
Pittsburgh PA, 15202

References

1 ResearchAndMarkets.com. (2024, October 2). Global $30 billion wound care market analysis and forecast to 2029: MiMedx and Smith & Nephew lead innovations [Press release]. https://www.globenewswire.com/news-release/2024/10/02/2956824/28124/en/Global-30-Billion-Wound-Care-Market-Analysis-and-Forecast-to-2029-MiMedx-and-Smith-Nephew-Lead-Innovations.html

2 Manna, B et al. (2024). Diabetes mellitus and wounds. In Evidence-based wound management. National Center for Biotechnology Information (U.S.). https://www.ncbi.nlm.nih.gov/books/NBK581940/

3 McKinsey & Company. (2024, April). Future of wellness: Trends. https://www.mckinsey.com/industries/consumer-packaged-goods/our-insights/future-of-wellness-trends

4 Team Dr. Joseph. (n.d.). Cooling leg relief gel 100ml. Retrieved from https://www.teamdrjoseph.com/en/cooling-leg-relief-gel-100ml/bc008

5 Kruze Consulting. (n.d.). Startup Payroll Costs: A Comprehensive Guide. 2025. Retrieved from https://kruzeconsulting.com/startup-payroll-costs/

6 Sen, C. K. (2019). Human wounds and its burden: An updated compendium of estimates. Advances in Wound Care, 8(2), 39–48.

7 Nussbaum, S. R., et. al. (2018). An economic evaluation of the impact, cost, and Medicare policy implications of chronic nonhealing wounds. Value in Health, 21(1), 27–32.

8 Grand View Research. (2023). Wound care market size, share & trends analysis report. Retrieved from https://www.grandviewresearch.com/industry-analysis/wound-care-market

9 International Diabetes Federation. (2021). IDF Diabetes Atlas (10th ed.). Brussels, Belgium: International Diabetes Federation. Retrieved from https://diabetesatlas.org

10 Armstrong, D. G. et al. (2017). Diabetic foot ulcers and their recurrence. The New England Journal of Medicine, 376(24), 2367–2375.

11 Zhang, P. et al. (2017). Global epidemiology of diabetic foot ulceration: a systematic review and meta-analysis. Annals of Medicine, 49(2), 106–116.

12 Centers for Disease Control and Prevention. (2020). National Diabetes Statistics Report 2020. Atlanta, GA: U.S. Department of Health and Human Services. Retrieved from https://www.cdc.gov/diabetes/data/statistics-report/index.html

13 Carinci, F., et. a;. (2020). Lower extremity amputations in persons with diabetes: a global perspective using data from the OECD and the QED study group. BMJ Quality & Safety, 29(4), 289–299.

14 Centers for Disease Control and Prevention. (2022). National Diabetes Statistics Report 2022. Atlanta, GA: U.S. Department of Health and Human Services. Retrieved from https://www.cdc.gov/diabetes/data/statistics-report/index.html

15 Rice, J. B. et al. (2014). Burden of diabetic foot ulcers for Medicare and private insurers. Diabetes Care, 37(3), 651–658. https://doi.org/10.2337/dc13-2176

16 Nussbaum, S. R. et. al. (2018). An economic evaluation of the impact, cost, and Medicare policy implications of chronic nonhealing wounds. Value in Health, 21(1), 27–32.

17 Grand View Research. (2023). Natural and organic personal care market size report. Retrieved from https://www.grandviewresearch.com/industry-analysis/natural-organic-personal-care-market

18 Allied Market Research. (2023). Topical pain relief market. Retrieved from https://www.alliedmarketresearch.com/topical-pain-relief-market

19 Tan, M., Chin, J. S., Madden, L. E., Knutsen, M. F., Ugland, H., Karlsson, M. K., Amiry-Moghaddam, M., & Becker, D. L. (2024). Oxygenated hydrogel promotes re-epithelialization and reduces inflammation in a perturbed wound model in rat. Journal of Pharmaceutical Sciences, 113(4), 999–1006.

20 Pandit, A. S.; Faldman, D. S., Effect of oxygen treatment and dressing oxygen permeability on wound healing. Wound Repair Regen 1994, 2 (2), 130-7.

21 IDF Diabetes Atlas 11th ed. (2025); chronic-wound burden (Sen 2025; Martinengo 2019); DFU epidemiology & outcomes (Zhang 2017; Armstrong NEJM 2017; Lo 2021 Singapore cohort; Jupiter/Armstrong JFAR 2020); World Bank/WHO; Grand View Research

22 11,000 clinics × 5 units/month + 11 hospitals × 24 units/month ≈ 663,000 units annually at *$15–25/unit*.

AethlBio